

MODULE IV

WATER SUPPLY

INTRODUCTION

- Next to air, for human life to exist, the important requirement is water
- Water is Nature's free gift to human race.
- Primary source of water is precipitation which falls on the earth's surface in the form of rain, snow, dew etc
- Rainfall is the most important part of precipitation
- The quantity and quality of water is important in a water supply scheme.

Sources for water supply

- surface sources
- Sub surface sources

Surface sources

- Lakes and streams
- Rivers
- Ponds
- Storage Reservoirs

Surface sources

- ***Lakes and streams*** : catchment area is small-quantity of water is very low –so not considered as primary source
- ***Rivers*** : Principal source of water supply schemes-excess water stored by constructing dams
- ***Ponds*** : A man made body of standing water smaller than a lake-water is used only for washing clothes or animals

Surface sources

- ***Storage Reservoir***

An artificial lake formed by the construction of dams across a river is termed as a storage reservoir.

main function is to stabilise the flow of water

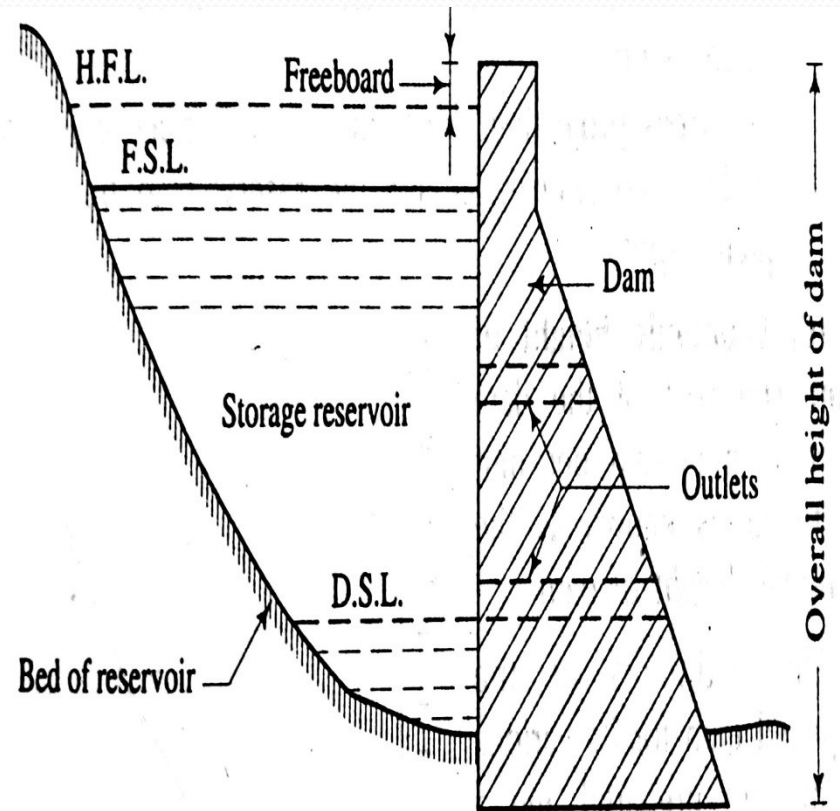
Storage Reservoir

- It consists of 3 parts

dam -to hold water

spillway -to allow the excess water to flow

gate chamber -to regulate the flow of water



Storage reservoir with dam

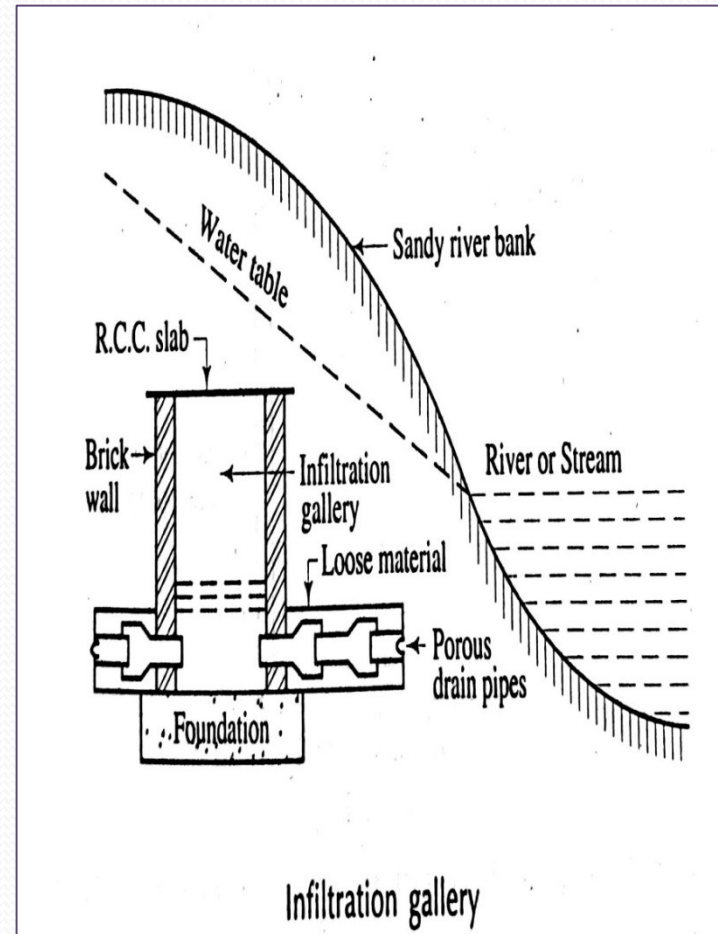
Sub surface/Under ground source

- Infiltration Galleries
- Infiltration wells
- Springs
- Wells

Sub surface/Under ground source

Infiltration galleries

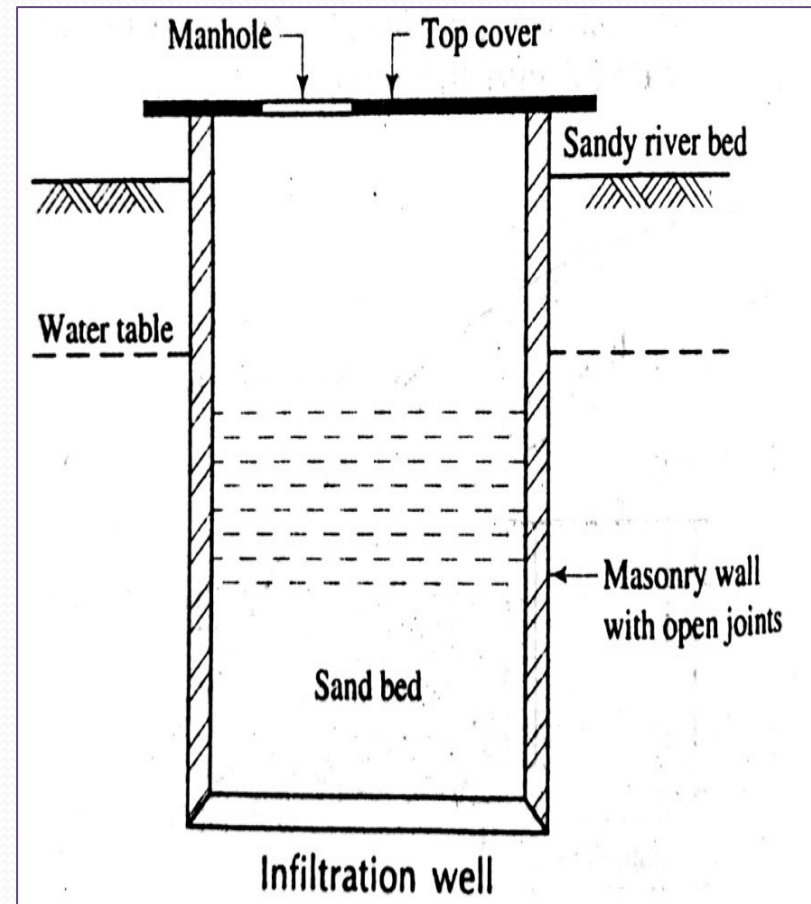
- horizontal tunnel.
- gets water from the water bearing strata by various porous drain pipes



Sub surface/Under ground source

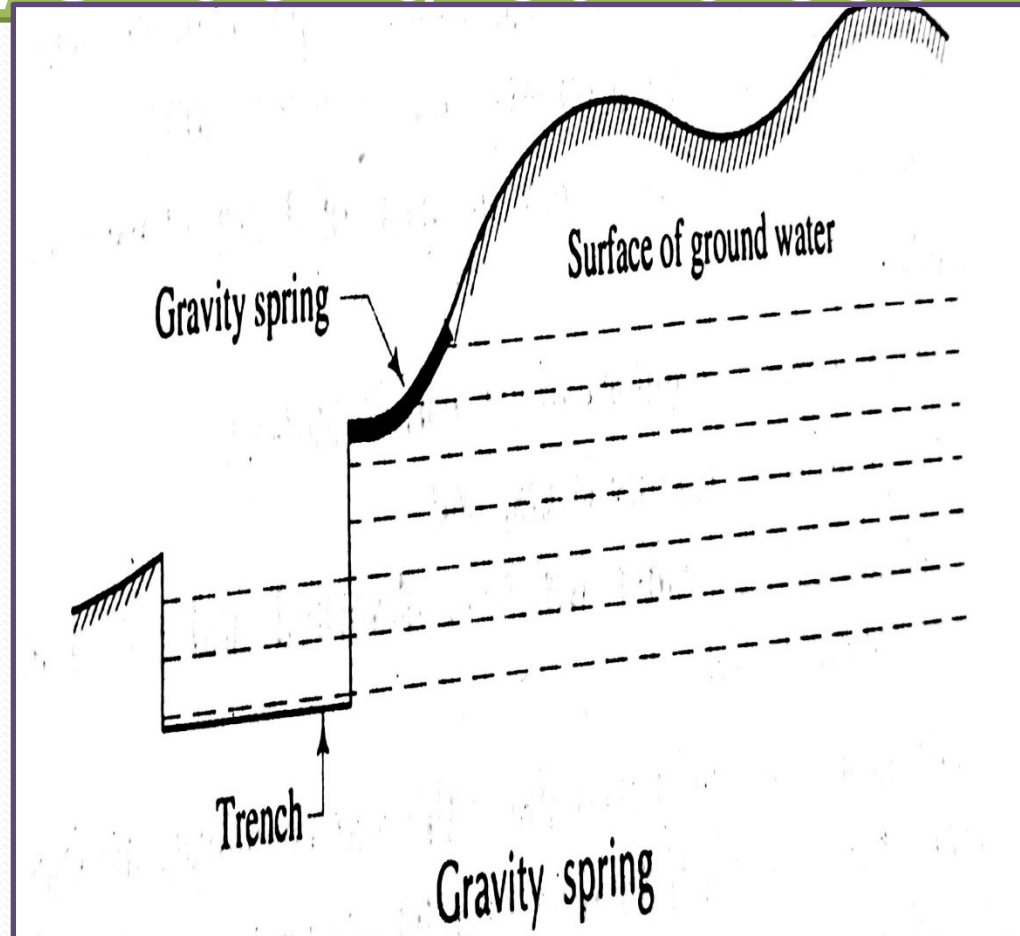
Infiltration wells

- In order to obtain large quantities of water the infiltration wells are sunk in series in the banks of river



Sub surface/Under ground source

- ***Springs-***
when
ground
water
appears
at the
surface,
springs are
formed



Sub surface/Under ground source

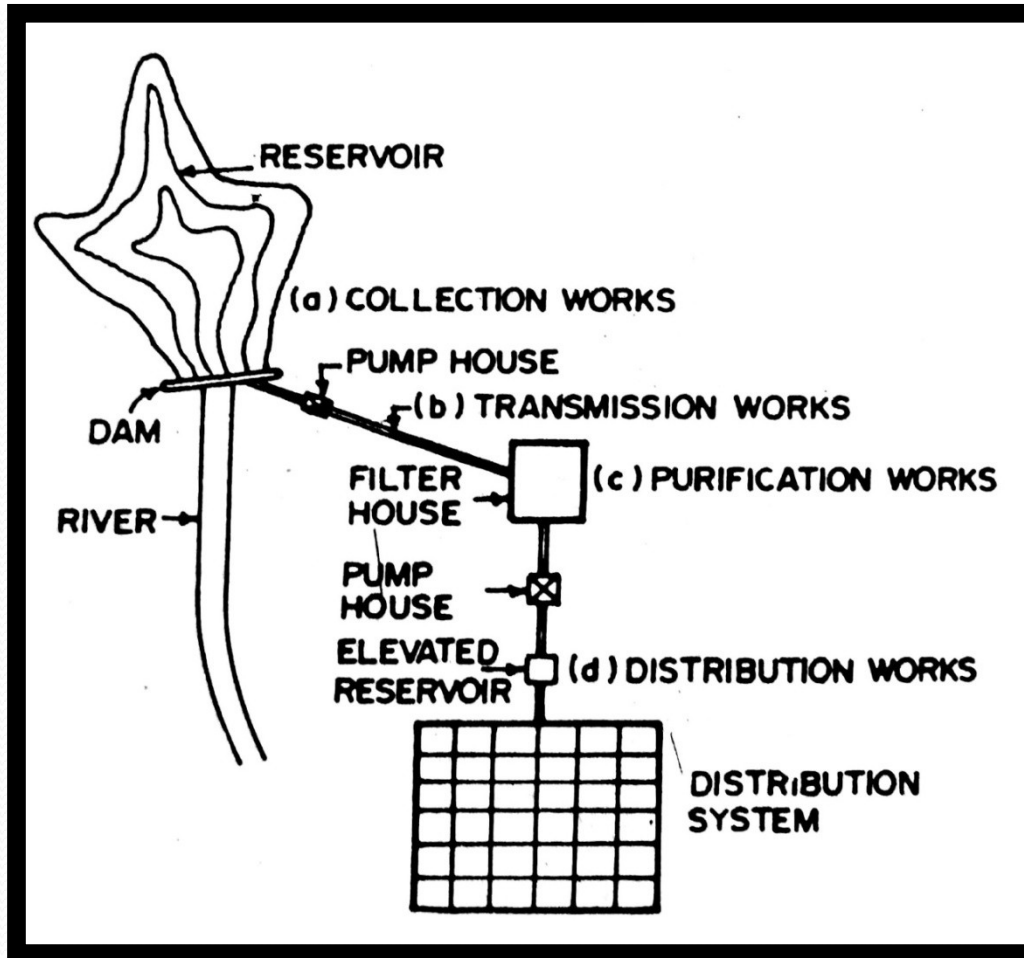
Wells- artificial hole or pit made in the ground for the purpose of tapping water.

Water supply systems

Municipal water systems consist of the following units

- ☐ Collection works
- ☐ Transmission works
- ☐ Purification works
- ☐ Distribution works

WATER SUPPLY SYSTEMS



Quality of water

- Pure water is not found in nature. It may contains impurirties like sand, silt, clay and other organic matter.
- Indian standard specification for drinking water
(IS 10500)

INDIAN STANDARD SPECIFICATIONS FOR DRINKING WATER

IS: 10500

S.NO.	Parameter	Requirement desirable Limit	Remarks
✓ 1.	Colour	5	May be extended up to 50 if toxic substances are suspected
✓ 2.	Turbidity	10	May be relaxed up to 25 in the absence of alternate
✓ 3.	pH	6.5 to 8.5	May be relaxed up to 9.2 in the absence
✓ 4.	Total Hardness	300	May be extended up to 600
5.	Calcium as Ca	75	May be extended up to 200
6.	Magnesium as Mg	30	May be extended up to 100
✓ 7.	Copper as Cu	0.05	May be relaxed up to 1.5
8.	Iron	0.3	May be extended up to 1
9.	Manganese	0.1	May be extended up to 0.5
✓ 10.	Chlorides	250	May be extended up to 1000
11.	Sulphates	150	May be extended up to 400
12.	Nitrates	45	No relaxation
✓ 13.	Fluoride	0.6 to 1.2	If the limit is below 0.6 water should be rejected. Max. Limit is extended to 1.5

14.	Phenols	0.001	May be relaxed up to 0.002
15.	Mercury	0.001	No relaxation
16.	Cadmium	0.01	No relaxation
17.	Selenium	0.01	No relaxation
✓ 18.	Arsenic	0.05	No relaxation
19.	Cyanide	0.05	No relaxation
✓ 20.	Lead	0.1	No relaxation
21.	Zinc	5.0	May be extended up to 10.0
22.	Anionic detergents (MBAS)	0.2	May be relaxed up to 1
23.	Chromium as Cr ⁺⁶	0.05	No relaxation
24.	Poly nuclear aromatic Hydrocarbons	--	--
25.	Mineral Oil	0.01	May be relaxed up to 0.03
26.	Residual free Chlorine	0.2	Applicable only when water is chlorinated
✓ 27.	Pesticides	Absent	--
28.	Radio active	--	--



THANK YOU